1. Clinical Question & Core Hypothesis

Clinical Question

Why do TLS-like lymphoid aggregates (histologic evidence of local immune organization) sometimes fail to confer progression-free benefit? The path suggests tumor-intrinsic PI3K signaling drives PD-L1 upregulation, which could neutralize TLS-mediated antitumor immunity. 2) Tier 1 paths used: (a) TMPRSS2/ERG cooperativity → PI3-kinase pathway activation; (b) PI3-kinase pathway activation → PD-L1 up-regulation. I built on these because they form a direct mechanistic axis linking oncogenic signaling to adaptive immune checkpoint expression. 3) Tier 2 supporting evidence: PSMA expression in stem-like tumor cells and the concept of cancer stem cell compartments provide a cellular reservoir that may sustain PI3K-driven, PD-L1-high clones resistant to immune control; androgen-deprivation/therapy-related selective pressures (from the path) offer a plausible context for selection of such signaling programs. 4) Endpoint choice: Progression-free interval (PFI) was chosen because the proposed mechanism (immune evasion via PD-L1 driven by PI3K) is expected to impact time to recurrence/progression more directly than long-term overall survival and because PFI is a robust, therapy-relevant clinical readout in TCGA-style cohorts. 5) Deliberate exclusions: I excluded BCG/urothelial-specific nodes, prostate-organ context, and broader carcinoma/general cancer-cell nodes as out-of-scope for a focused mechanistic test in epithelial tumors; I also limited the hypothesis to one imaging feature and two core molecular anchors to maintain mechanistic specificity?

Working Hypothesis

Mechanistic hypothesis: In breast cancer, the presence of TLS-like lymphoid aggregates is associated with longer progression-free interval (PFI) only when tumor cells do not exhibit PI3K pathway-driven PD-L1 upregulation. When PI3K pathway activation (represented by PIK3CA alterations or expression-based PI3K activity signatures) drives upregulation of PD-L1 (CD274) on tumor cells, TLS-like aggregates fail to confer TLS-associated anti-tumor immunity, resulting in shorter PFI. Therefore, TLS status, PD-L1 expression, and PI3K activation interact to shape the prognosis of breast tumors, with the strongest adverse prognosis observed in TLS-high, PD-L1-high, PI3K-active tumors compared with TLS-high, PD-L1-low or TLS-low tumors. Secondary modifiers TMPRSS2/ERG and PSMA signatures may modulate this axis as contextual factors but are not required anchors; they could reflect broader oncogenic signaling cross-talk that reinforces PD-L1 upregulation or immune exclusion in certain subsets. Experimental scope: (i) test the three-way interaction TLS × PD-L1 × PI3K on PFI using TCGA-BRCA data; (ii) validate TLS detection and PD-L1/PI3K activity readouts with orthogonal measures; (iii) exploratory analysis of TMPRSS2/ERG and PSMA signatures as potential modulators; (iv) if feasible, perform targeted in vitro assays to assess PD-L1 induction by PI3K pathway activation and its impact on T cell–mediated cytotoxicity in TLS-mimetic contexts.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	PI3-kinase pathway activation, PD-L1
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

PI3-kinase pathway activation: PI3-kinase pathway activation; (b) PI3-kinase pathway activation → PD-L1 up-regulation.

PD-L1: PD-L1 driven by PI3K) is expected to impact time to recurrence/progression more directly than long-term overall survival and because PFI is a robust, therapy-relevant clinical readout in TCGA-style cohorts.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	TMPRSS2/ERG	Supporting modifier from knowledge graph
Secondary	PSMA	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- TMPRSS2/ERG: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- PSMA: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.